# OpenManipulator-X

ISO 9283 Certificate

**Certificate ID:** AC-RUN_001

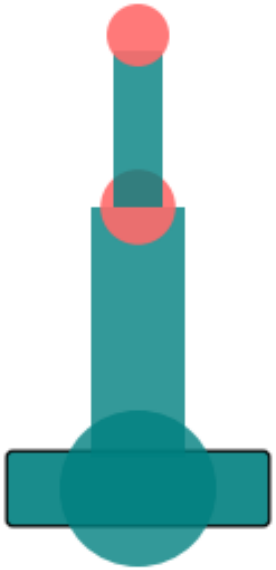
**Issue Date:** 2026-04-07

**Laboratory:** AutoCert Lab

**Valid Until:** 2027-04-07

**Technician:** AutoCert System

## Robot Model



## General Specification

Payload: 0.3 kg

Reach: 485 mm

Degrees of freedom: 5

Programming: ROS 2 + MoveIt 2

Material: PLA / Aluminum

Weight: 0.63 kg

## Joint Specifications

| Axis |
| --- |
| Range |
| Speed |
| Joint 1 |
| ±180° |
| 275°/s |
| Joint 2 |
| ±86° |
| 275°/s |
| Joint 3 |
| -86° to 80° |
| 275°/s |
| Joint 4 |
| -97° to 113° |
| 275°/s |
| Joint 5 |
| ±180° |
| 275°/s |

Input voltage: 12 V

Power consumption: 35 W

Maximum power: 120 W

IP Rating: IP20

Noise: $<$ 60 dB(A)

Communication: TTL Dynamixel bus / ROS 2

## Power & System

# OpenManipulator-X

ISO 9283 Certification Report

## Certification Results

Mean Accuracy: **0.456 mm**

Max Accuracy: **0.789 mm**

Mean Repeatability: **0.089 mm**

Max Repeatability: **0.145 mm**

Orientation Accuracy: **0.342 deg**

Status: **FAIL**

## Movement Performance

Max TCP Speed: 0.3 m/s

Max Joint Speed: 275 °/s

Pose Repeatability: ± 0.20 mm

## Workspace Analysis

Max Radius: 0.314 m

Min Radius: 0.280 m

Workspace Volume: 0.0806 $m^3$

Surface Area: 0.919 $m^2$

## Reachability Metrics

Workspace Coverage: 1.0%

Point Density: 0.557

Dexterous Volume: 0.0388 $m^3$

## Physical Characteristics

Footprint: Ø150 mm

Tool Flange: Custom interface

Connector: TTL bus

Cable Length: 1 m

## Test Conditions

Temperature: 22.5°C

Humidity: 45%

Measurement System: Leica AT960

Calibration Date: 2024-01-15

Test Cube Size: 50 mm

Repetitions: 5

## Certification Authority

Organization: AutoCert Lab

Phone: +91-9876543210

Email: autocert@robotics.com

Website: www.autocert.com

Country: India

This certificate is electronically generated and valid without signature.